Olonade Kelvin

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Aspiring RTL/VLSI Design and Verification Engineer with hands-on silicon experience via Tiny Tapeout (SkyWater 130nm), strong RTL design and verification fundamentals, and a track record of building IP cores from spec to signoff. Founding chair of the first IEEE Solid-State Circuits Society chapter at UNILAG. Passionate about digital design, low-power circuits, and tape-out-ready workflows.

Education

University of Lagos — Lagos, Nigeria
B.Sc. Electrical & Electronics Engineering | CGPA: 4.18/5.00 — *Oct 2023 – Oct 2028*

Relevant Coursework: Digital Logic Design, Electronic Circuits, Physical Electronics, Signals & Systems.

Technical Skills

HDL & Design: Verilog, SystemVerilog, RTL Design, FSM Design, Synthesizable IP, Parameterized Modules
Verification: Self-checking Testbenches, Functional Coverage, Constrained-Random, Assertions, Edge-Case Analysis
EDA & Flow: OpenLane, Yosys, Magic, KLayout, Icarus Verilog, GTKWave, Verilator, OpenROAD, Cocotb
Process & PDKs: SkyWater SKY130, Standard-Cell Synthesis, STA Basics, Place & Route, DRC/LVS
Embedded/Tools: PIC16F (CLB/CIP), ARM Cortex-M, C/C++, Python, Git, Linux, MATLAB/Simulink

Silicon & Hardware Projects

Autonomous VGA Algorithmic Synthesizer ASIC | Verilog, SKY130, OpenLane

- Designing a tape-out-ready digital ASIC for **Tiny Tapeout SKU26A** on **SkyWater 130nm**, generating autonomous VGA 640×480 visuals from on-chip fractal and LFSR engines—**zero firmware, pure digital logic**.
- Hand-optimizing RTL for extreme area constraints (1 tile $\approx 160\times100\,\mu m$); pipelining pixel pipeline at 25.175 MHz and balancing critical paths through OpenLane synthesis and STA.
- Implementing self-checking SystemVerilog testbench with functional coverage on sync timing, fractal correctness, and entropy quality before signoff.

Synthesizable UART TX/RX IP Core | Verilog, Cocotb, Verilator

- Designed a fully parameterized UART IP (baud, data bits, parity, stop bits) achieving **100% functional coverage** via a self-checking testbench with constrained-random stimulus and golden-model reference.
- Verified back-to-back transmission, framing/parity error injection, and metastability handling across asynchronous domains using two-flop synchronizers.

Hardware EEPROM Wear-Leveling Controller | Verilog, SKY130, Tiny Tapeout

- Designed and taped out a compact, attack-resistant wear-leveling controller for external EEPROM/flash memories, implementing the **Start-Gap algorithm** (used in 3D XPoint) to dramatically extend memory lifetime. Verified with cocotb and synthesized on SkyWater 130nm via Tiny Tapeout SKU26A.
- Incorporated a **2-round Feistel address scrambler** with LFSR key to defeat targeted wear attacks, saturating 8-bit per-block wear counters, automatic bad-block retirement, and in-band telemetry for wear statistics (max-min skew, total write count).
- Achieved successful GDS generation with place-and-route; passed full functional test suite in simulation, proving a fully synthesizable, silicon-ready IP core with hardware-level security and reliability.

Thermal Modeling of a Multi-Core Processor Die | MATLAB, Numerical Methods

- Built a 2D transient finite-difference thermal solver to analyze **hotspot evolution and thermal crosstalk** between adjacent cores under realistic power maps.
- Quantified peak junction temperature reduction via floorplan permutation, informing power-aware placement decisions.

Urban FloodGuard — Firmware-Free Flood Monitor | PIC16F CLB, Pure Hardware

- Implemented a deterministic flood-alert system using **Configurable Logic Blocks and Core Independent Peripherals only**—no MCU code—demonstrating mastery of pure combinational/sequential design on silicon.

Experience

Embedded Systems Engineer — May 2026 – Present
Balancee Tech Solutions — *Ajah city, Lagos*

- Leading end-to-end design of a custom fuel-pump embedded controller: schematic, MCU firmware, sensor signal chain, and field-test verification.

Embedded Systems Intern — Dec 2025 – Present
AVZDAX — *Victorial Island, Lagos*

- Delivered an edge computer-vision pipeline on Raspberry Pi using hardware-software co-design, optimizing for latency and memory footprint on a constrained SoC.

Leadership & Certifications

Founding Chair — IEEE Solid-State Circuits Society (SSCS) Student Chapter, University of Lagos: establishing the **first SSCS chapter in the institution**, organizing RTL/ASIC workshops, and building Nigeria's pipeline of silicon engineers.
Arm Developer Program Contributor (Nov 2025 – Present) — contributing technical content on Cortex-M architectures.
Certifications: Zero to ASIC Course (2026), ARM Cortex-M Processors Overview, Logic Circuit Design 101 (Microchip), Rapid Prototyping with Curiosity Nano, Advanced Embedded C, UC Irvine IoT & Embedded Systems, Modeling & Simulation with Simulink.